

Technical Task

Product Recommendation and Customer Targeting Analysis

Context

You are working with customer and product recommendation data for an e-commerce company. The business wants to understand which products, brands, categories, and customer segments have the strongest recommendation potential. Your task is to clean, analyze, and prepare this data using Python.

Datasets

You have two datasets:

- customer_data_collection.csv
- product_recommendation_data.csv

Customer dataset fields

Field	Description / Use
Customer_ID	Unique customer identifier
Age	Customer age
Gender	Customer gender
Location	Customer geographic location
Browsing_History	Products or categories viewed by customer
Purchase_History	Products or categories previously purchased
Customer_Segment	Customer segment classification
Avg_Order_Value	Average order value for the customer
Holiday	Holiday context associated with the customer record
Season	Season associated with the customer record

Product dataset fields

Field	Description / Use
Product_ID	Unique product identifier
Category	Product category
Subcategory	Product subcategory
Price	Product price
Brand	Product brand
Average_Rating_of_Similar_Products	Benchmark rating for similar products
Product_Rating	Rating of the product
Customer_Review_Sentiment_Score	Review sentiment score
Holiday	Holiday context associated with product recommendation
Season	Season associated with product recommendation
Geographical_Location	Location where product recommendation is relevant
Similar_Product_List	Related or similar products
Probability_of_Recommendation	Model or business probability that product should be recommended

Tasks

1. Load and inspect the datasets

- Load both CSV files using pandas.
- Check shape, column names, data types, missing values, duplicate rows, and first few rows.
- Remove unnecessary Unnamed columns if present.

2. Clean the data

- Remove duplicate rows.
- Handle missing values using clear and explainable rules.
- Check invalid ages; age should be positive and realistic.
- Check invalid prices; price should be greater than 0.
- Check invalid product ratings; ratings should be between 0 and 5.
- Check invalid average ratings of similar products; ratings should be between 0 and 5.
- Check customer review sentiment score is within a valid range.
- Check probability of recommendation is between 0 and 1.
- Standardise relevant text columns such as brand, category, location, holiday, and season.

3. Create customer-level analysis

- Number of customers per segment.
- Average order value per customer segment.
- Average order value per location.
- Customer count by gender.
- Customer count by season and holiday.
- Answer: Which customer segment has the highest average order value?
- Answer: Which location has the highest average order value?
- Answer: Which season has the highest-value customers?

4. Create product-level analysis

- Average product price by category.
- Average product rating by category.
- Average recommendation probability by category.
- Average recommendation probability by brand.
- Average sentiment score by brand.
- Number of products per category.
- Answer: Which category has the highest recommendation probability?
- Answer: Which brand has the strongest product rating?
- Answer: Which brand has the best sentiment score?
- Answer: Which category has the highest average price?

5. Analyse recommendation potential

- Create a product recommendation score table by brand and category.
- For each brand and category, calculate average product rating, average sentiment score, average recommendation probability, average price, and number of products.

- Define high-potential products using this rule: Product_Rating ≥ 4 , Customer_Review_Sentiment_Score $\geq$ dataset average, and Probability_of_Recommendation $\geq$ dataset average.
- Create a high-potential products table with Product_ID, Brand, Category, Subcategory, Price, Product_Rating, Customer_Review_Sentiment_Score, Probability_of_Recommendation, Season, Holiday, and Geographical_Location.

6. Match products to customer segments

- Create a basic recommendation targeting table.
- Match customers to products where customer Location equals product Geographical_Location, customer Season equals product Season, and customer Holiday equals product Holiday.
- For each matched customer-product pair, include customer details and product details.
- Add recommendation_type = high_priority if Probability_of_Recommendation ≥ 0.75 , otherwise standard.

7. Campaign recommendation

- For each brand, recommend the best customer segment to target.
- Use this rule: Best segment = segment with the highest average order value among customers matched to that brand.
- Create a final table with Brand, Best_Customer_Segment, Average_Order_Value, Number_of_Customers, Average_Recommendation_Probability, Best_Location, and Best_Season.

8. Create visualisations

- Create a bar chart: Average recommendation probability by product category.
- Create a bar chart: Average order value by customer segment.
- Optional: Create a scatter plot showing product rating vs recommendation probability.

9. Export outputs

- Export high_potential_products.csv.
- Export brand_campaign_recommendations.csv.
- Optional: Export customer_product_recommendation_matches.csv.
- Optional: Export final_analysis_outputs.xlsx with multiple sheets.

Deliverables

- Jupyter Notebook or Python script
- Cleaned datasets
- Final aggregated CSV or Excel outputs
- Charts
- Short explanation of findings

Suggested final business questions to answer

1. Which product category has the highest recommendation probability?
2. Which brand should be prioritised for marketing?
3. Which customer segment has the highest average order value?
4. Which location has the strongest customer value?

5. Which products should be promoted first?
6. Which customer segments should receive targeted recommendations?

Notes

Clean, modular code is encouraged. Use functions or `pandas.pipe` where appropriate. Clearly explain cleaning decisions and business rules used for recommendation and campaign targeting.